Verification of a Conjugate Heat Transfer Model in OpenFOAM for a Simplified Medical Device Geometry

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Computational models and simulations (CM&S) are increasingly used in medical device development, yet their adoption in regulatory decision-making remains limited due to challenges in establishing model credibility. This study presents a structured and reproducible verification framework for a computational model of a simplified medical device involving conjugate heat transfer using `OpenFOAM`*'s* `chtMultiRegionFoam` *solver. Code verification is performed using analytical benchmarks for laminar pipe flow, radial heat conduction, and transient thermal diffusion yielding observed orders of convergence of* $p = 1.95$*,* $p = 2.08$*, and* $p = 2.15$ *(at 0.025 s) respectively. Solution verification evaluates discretization error through systematic mesh and timestep refinement and regression-based extrapolation using the* `StREEQ` *method. Results show that spatial and temporal discretization errors in transient coupled simulations interact and cannot be assessed independently. Co-refinement is required to capture true convergence behavior, while independent refinement yields polluted convergence rates. Computational cost analysis demonstrates diminishing returns with refinement and highlights that runtime does not vary monotonically with discretization parameters due to solver convergence behavior. The use of a simplified geometry enables rigorous verification while retaining the dominant physics of the system. Grounded in a risk-informed framework consistent with ASME V&V 40 Standard, this study demonstrates that a level of verification rigor beyond minimal practice can be achieved through targeted model simplification and systematic analysis under realistic computational constraints.*

Keywords: verification and validation, solution verification, conjugate heat transfer, computational fluid dynamics, medical device modeling, model credibility, reproducibility, discretization error, OpenFOAM, V&V 40

1 Introduction

Computational models and simulations (CM&S) are increasingly used across the medical device lifecycle, from early stages in design to optimization of in silico testing and post-market surveillance [1]. These tools offer cost-effective, flexible, and scalable alternatives to traditional experimentation. However, their use in high risk regulatory decision making remains limited [2, 3]. To be accepted by regulatory agencies such as the U.S. Food and Drug Administration (FDA), computational models must be supported by rigorous, transparent evidence demonstrating that they are appropriate for their use case and capable of producing trustworthy results.

Establishing this trust is particularly challenging for systems involving multiphysics interactions, coupled domains, and inherent uncertainties, especially when experimental data are sparse, costly, or difficult to obtain. In such settings, verification, validation, and uncertainty quantification (VVUQ) are essential. However, the required level of rigor must be aligned with the model's intended use and associated risk. For medium to low risk applications, comprehensive experimental studies or high-fidelity simulations may be impractical. In these cases, credibility depends not only on numerical rigor but also on transparency, including clearly communicating assumptions, limitations, and modeling decisions to support risk-informed interpretation.

To provide context for the level of verification performed, this study adopts a risk-informed perspective consistent with the ASME V&V 40 framework. For the present application, the model is characterized by a low decision consequence and moderate model influence, corresponding to a medium–low risk classification. Under this classification, a moderate to minimal level of verification would typically be sufficient. However, in this work, additional rigor is pursued where computationally feasible. In particular, simplified geometries and isolated physical configurations are used to enable more detailed assessment of numerical behavior, including estimation of discretization error and evaluation of observed order of accuracy. This approach provides a level of verification beyond minimal while remaining practical.

This study focuses specifically on verification as a foundational component of the VVUQ process, with validation and uncertainty quantification addressed in a subsequent companion study. In particular, this study presents a structured and transparent verification study of a computational model for a medical device application involving heated internal flow. The system is modeled as a conjugate heat transfer (CHT) problem, where laminar fluid flows through a heated solid casing and exchanges thermal energy before exiting the device. Simulations are performed using the open-source finite volume solver `OpenFOAM`, specifically the `chtMultiRegionFoam` solver for coupled fluid-solid heat transfer [4, 5].

The verification approach focuses on both code and solution verification to establish numerical correctness and quantify discretization error. This includes the use of analytical problems, systematic mesh and timestep refinement studies, and regression-based error estimation. Particular emphasis is placed on transient, coupled configurations, where spatial and temporal resolution interact and

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cannot be assessed independently.

This work is grounded in the ASME V&V 40 framework for credibility assessment in medical device applications and aligns with FDA-recognized standards, including V&V 10, V&V 20, and V&V 40 [6–13]. While these standards define high-level credibility principles, they do not prescribe specific verification workflows. To address this gap, the methodology presented here draws on best practices from high-consequence fields such as aerospace and nuclear engineering, where VVUQ processes are well established due to similar experimental and modeling constraints [14–16]. Additional details on the risk-informed credibility assessment, including V&V 40 gradation selection and justification, are provided in publicly accessible repository materials [17].

By openly documenting simulation workflows, input files, and post-processing procedures, this study provides a reproducible case study in verification for medical device modeling. The goal is to demonstrate how rigorous, transparent, and risk-informed verification can be achieved within realistic computational and resource constraints, supporting broader adoption of CM&S in regulatory and engineering applications.

2 Methods

This study is motivated by a heat transfer driven electronic drug delivery system (EDDS) medical device, which uses applied power to heat fluid flowing through the device and into a users mouth cavity. While the full system includes complex components such as coils for heating and an expanding mouthpiece, verifying the entire system limits the level of rigor achievable in code and solution verification due to geometric and multiphysics complexity.

To enable systematic verification, a simplified geometry is adopted that preserves the dominant transport mechanisms of laminar flow and conjugate heat transfer while minimizing geometric complexity. This reduction enables direct comparison to analytical solutions and supports high-resolution studies. By verifying key physical processes in isolation, this approach establishes confidence in the numerical modeling framework and provides a foundation for progressively reintroducing complexity in future validation and application studies.

2.1 Verification Framework. Verification is divided into two categories: *code verification*, which assesses the correctness of the numerical implementation through comparison with analytical solutions, and *solution verification*, which quantifies numerical uncertainty arising from spatial and temporal discretizations, solver settings, and modeling assumptions. These activities follow the framework outlined in ASME V&V 20 and V&V 40 [7, 8], and support a structured credibility assessment of simulation results for subsequent validation and decision-making.

To isolate sources of numerical error, each verification component is further subdivided according to the underlying physical processes and modeling complexity. Code verification is performed using two cases: hydrodynamic flow and radial heat conduction, which independently assess the solver's ability to reproduce fluid and thermal transport behavior.

Solution verification is conducted using coupled fluid and thermal configurations which increase in complexity. These include a steady-state coupled pipe model and a transient, device-representative EDDS configuration. The latter provides the closest approximation to the target medical device and enables evaluation of numerical uncertainty under application relevant conditions.

2.2 System Description & Operating Conditions. The pipe system is designed to approximate the geometry of the medical device, including both the internal fluid channel and surrounding casing. The fluid domain is partitioned into three regions: the pipe inlet, the pipe case, and the pipe outlet. The central pipe section, shown in Figure 1, is thermally coupled to a surrounding solid domain representing the casing, within which volumetric heat generation is applied.

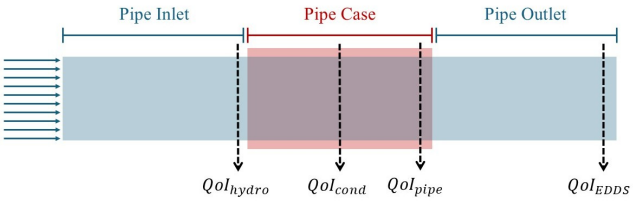


Fig. 1 Schematic of the generalized pipe configuration used across all verification studies.

The pipe has an internal diameter of 4.54 mm and is surrounded by a casing with a thickness of 0.75 mm. Operating conditions impose a total heating power of 1 W, applied uniformly as a volumetric heat source within the casing for a duration of 10 seconds. The fluid enters with a uniform velocity of 0.5 m/s and an initial temperature of 20°C, resulting in a laminar, transient conjugate heat transfer problem involving coupled solid-fluid thermal interactions.

All verification studies introduce controlled variations in geometry, temporal regime, and quantities of interest, as summarized in Table 1. For code verification, the configuration is intentionally simplified to isolate numerical behavior and enable comparison with analytical solutions. A sufficiently long pipe is employed to ensure fully developed hydrodynamic and thermal conditions, thereby minimizing inlet and outlet effects.

To achieve this, the pipe length is selected to exceed both the hydrodynamic and thermal entry lengths, given by

$$L_{\text{hydro}} = 0.05\, Re\, D \approx 36 \text{ mm}, \quad L_{\text{therm}} = 0.05\, Re\, Pr\, D \approx 25 \text{ mm} \tag{1}$$

where Re is the Reynolds number, Pr is the Prandtl number, and D is the pipe diameter. These conditions ensure fully developed flow and thermal fields and support rigorous assessment of discretization accuracy under steady-state assumptions.

In contrast, the solution verification cases are designed to more closely represent the medical device configuration. These cases incorporate increased geometric complexity and are modeled under transient conditions to capture time-dependent system behavior and assess numerical uncertainty under application relevant conditions.

2.3 Governing Equations & Numerical Approach. Simulations were performed using the open-source finite-volume solver OpenFOAM. Conjugate heat transfer between the solid casing and fluid domain is modeled using the `chtMultiRegionFoam` solver, which solves the incompressible Navier-Stokes equations in the fluid region and transient heat conduction in both fluid and solid domains without phase change. Conjugate coupling is enforced at the fluid-solid interface through continuity of temperature and heat flux. All equations are discretized using the finite-volume method on unstructured hexahedral meshes.

Fluid Region. The fluid domain (air) is modeled as an incompressible, Newtonian fluid with constant properties under laminar flow conditions. The governing equations include conservation of mass (Eq. 2), conservation of momentum (Eq. 3), and thermal energy transport (Eq. 4).

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0 \tag{2}$$

$$\frac{\partial (\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U}\mathbf{U}) = -\nabla p + \nabla \cdot \tau + \rho g \tag{3}$$

$$\frac{\partial (\rho c_p T)}{\partial t} + \nabla \cdot (\rho c_p \mathbf{U} T) = \nabla \cdot (\kappa \nabla T) + S_E \tag{4}$$

where $\mathbf{U}$ is the fluid velocity, T is the temperature, t is time, ρ is density, p is pressure, $\mathbf{g}$ is gravitational acceleration, τ is the shear stress, c_p is specific heat capacity, κ is thermal conductivity, and S_E is the volumetric heat source [18, 19].

Table 1 Geometry quantities, in mm, for the code and solution verification studies. The pipe diameter is 4.54 mm and the case thickness is 0.75 mm.

Model	Verification Study	Regime	Geometry	pipe inlet	pipe case	pipe outlet	pipe	case
Hydrodynamic	Code	Steady	long pipe	45	45	0	90	45
Radial Conduction	Code	Steady	medium pipe	11.25	45	0	56.25	45
Thermal Diffusion	Code	Transient	medium pipe	11.25	45	0	56.25	45
Coupled Pipe	Solution	Transient	long pipe	45	45	0	90	45
Simplified EDDS	Solution	Transient	short pipe	2.021	7.2	42.509	51.73	7.2

Solid Region. The solid casing (nickel) is modeled as an isotropic material with constant thermal properties. Transient heat conduction in the solid is governed by

$$\frac{\partial(\rho c_p T)}{\partial t} = \nabla \cdot (\kappa \nabla T) + S_E \tag{5}$$

The variables ρ, c_p, κ, and S_E retain the same physical interpretations as in the fluid region. At the fluid-solid interface, continuity of temperature and heat flux is enforced to ensure conjugate heat transfer.

Material Properties. Thermophysical properties used for air (fluid) and nickel (solid casing) are summarized in Table 2, consistent with values at 20°C.

Table 2 Material properties used in simulations.

Property	Symbol	Air (fluid)	Nickel (solid)
Density	ρ [kg/m^3]	1.204	8908
Specific Heat	c_p [J/(kg·K)]	1007	444
Thermal conductivity	κ [W/(m·K)]	0.027	90.9
Viscosity	μ [Pa·s]	1.48×10^{-5}	—

Boundary and Initial Conditions (B&IC). Boundary and initial conditions are defined to isolate the governing physical processes for each verification configuration. For code verification, simplified conditions are applied to enable comparison with analytical solutions. The hydrodynamic case assumes steady, isothermal flow with a uniform inlet velocity and fixed outlet pressure, while all walls are maintained at a constant temperature. The radial conduction case applies a uniform volumetric heat source within the solid domain with specified temperature at the outer boundary and an adiabatic interface to isolate conduction.

For solution verification, boundary conditions reflect coupled fluid-thermal behavior. The fluid enters with a uniform velocity of 0.5 m/s and temperature of 20°C, with outlet pressure fixed at 101.325 kPa and zero-gradient conditions applied to velocity and temperature. No-slip conditions are applied at all walls, and conjugate heat transfer is enforced at the fluid-solid interface. For transient simulations, both domains are initialized at 20°C, with the fluid initially at rest, and a volumetric heat source is applied within the solid casing for a specified duration.

Numerical Discretization. A second-order backward scheme is used for temporal discretization. Spatial discretization employs Gauss linear schemes for gradient and diffusion terms. Convective fluxes are discretized using an upwind scheme, with velocity gradients treated using Gauss upwind.

Solver Configuration. The `chtMultiRegionFoam` solver employs a segregated solution strategy, in which the governing equations for fluid flow and heat transfer are solved iteratively. Pressure-velocity coupling is handled using the PIMPLE algorithm [4, 5].

Parallel Implementation. All simulations are executed in parallel using domain decomposition. Mesh partitioning is performed using the hierarchical decomposition method, and results are reconstructed using `reconstructPar` [20–22].

2.4 Quantities of Interest & Error Metrics. Quantities of interest (QoIs) are defined to isolate error contributions associated with each verification study (see Figure 1). To isolate hydrodynamic error, QoI_{hydro} is defined at 1.25 times the hydrodynamic entry length, ensuring fully developed flow conditions. To assess thermal conduction while minimizing boundary effects, QoI_{cond} is defined at the thermal entry length. To assess the transient thermal diffusion, QoI_{diff} is defined 1.2 times the thermal entry length. The steady-state coupled pipe configuration is evaluated at QoI_{pipe}, defined as 1.2 times the thermal entry length. For the transient, device-representative case, QoI_{EDDS} is defined 1 mm upstream of the outlet to capture application-relevant behavior.

For code verification with a known analytical solution, the analytical solution is evaluated at the same spatial cross sections as the numerical solution. Pointwise comparisons are performed using the normalized L_2 error, defined as:

$$L_2 = \left(\frac{\sum_i (\phi_{i,\text{num}} - \phi_{i,\text{analytic}})^2 \text{V}_i}{\sum_i (\phi_{i,\text{analytic}})^2 \text{V}_i} \right)^{1/2} \tag{6}$$

where $\phi_{i,\text{num}}$ and $\phi_{i,\text{analytic}}$ represent the numerical and analytical values of the QoI at the ith centroid and vol_i is the ith centroid volume.

For solution verification, QoIs are compared using the relative error, defined as

$$Err_{\text{relative}} = \frac{|\phi_{\text{num}} - \phi_{\text{StREEQ}}|}{\phi_{\text{StREEQ}}} \tag{7}$$

where ϕ_{StREEQ} is the extrapolated QoI obtained from the StREEQ regression method.

2.5 Verification Study Design. Verification studies are designed to isolate sources of numerical error and assess solver performance across increasing levels of physical and geometric complexity. The study is divided into code verification and solution verification, followed by resolution studies used to quantify discretization error and assess solver sensitivity.

2.5.1 Code Verification Studies. Code verification evaluates whether the numerical implementation correctly solves the governing equations by comparison with analytical solutions. These studies are conducted in simplified configurations to isolate individual physical processes.

Hydrodynamic (Fluid Only) Model. This configuration isolates fully developed laminar pipe flow (Hagen-Poiseuille flow) under steady, isothermal conditions. Heat transfer is neglected, and the pipe length is set to exceed the hydrodynamic entry length to ensure fully developed flow. The fluid enters with a uniform velocity of 0.5 m/s and temperature of 20°C, with outlet pressure

fixed at 101.325 kPa. All walls enforce no-slip conditions and are maintained at constant temperature. The QoI, QoI_{hydro}, is the axial velocity profile at a cross section located within the fully developed region, enabling direct comparison with the analytical solution. Spatial convergence is assessed using four progressively refined meshes with representative cell sizes of 0.43 mm, 0.22 mm, 0.11 mm, and 0.088 mm.

Radial Conduction Model. This configuration isolates steady-state heat conduction in the solid domain. Fluid effects are removed by imposing an adiabatic boundary at the fluid-solid interface. A uniform volumetric heat source of 18 Watts is applied within the casing, and the outer wall is maintained at a fixed temperature, 293.15 K. The QoI, QoI_{cond}, is the radial temperature profile at the midpoint of the casing, sufficiently far from boundary effects. Spatial convergence is assessed using three progressively refined meshes with representative cell sizes of $1.4e^{-4}$ m, $7.1e^{-5}$ m, and $3.5e^{-5}$ m.

Transient Thermal Diffusion Model. This configuration isolates transient thermal diffusion in the fluid domain by applying a zero fluid velocity, eliminating advective transport effects. The governing equation reduces to a transient diffusion problem and an analytical solution is obtained through separation of variables with Bessel functions. The fluid domain is initialized at 20°C, while the solid domain is initialized at 30°C. The QoI, QoI_{diff}, is the transient radial temperature profile extracted at a cross section in the fluid region at 1.2 times the case region. Spatial and temporal convergence are assessed using three co-refined meshes and time-steps with representative cell sizes of $1.3e^{-4}$ m, $6.5e^{-5}$ m, and $3.2e^{-5}$ m and co-refined time steps of $5e^{-3}$ s,$5e^{-4}$ s, and $5e^{-5}$ s.

2.5.2 Solution Verification Studies. Solution verification assesses numerical uncertainty in coupled fluid-thermal simulations without available analytical solutions. The study uses a set of seven mesh resolutions with corresponding characteristic cell sizes, paired with decreasing time steps to assess both spatial and temporal convergence. This range of resolutions enables evaluation of discretization error under conditions representative of the coupled fluid-thermal system.

Steady-State Coupled Pipe. This configuration introduces steady-state conjugate heat transfer under laminar flow. The fluid inlet velocity and outlet pressure conditions remain consistent with the hydrodynamic case. A constant temperature, 50°C, is imposed within the solid region to drive thermal development. The QoI, QoI_{pipe}, is the bulk fluid temperature at a downstream location beyond the thermally developing region. Discretization error is quantified through mesh refinement and relative error with respect to extrapolated solutions. Spatial convergence is assessed using the same set of four progressively refined meshes employed in the code verification studies, ensuring consistency with the steady-state analytical configurations.

Transient Simplified EDDS. This configuration represents a device-relevant geometry under transient conjugate heat transfer. The fluid enters at 0.5 m/s and 20°C, with pressure outlet conditions and no-slip walls. A volumetric heat source is applied within the solid casing for a finite duration. The QoI, QoI_{EDDS}, is the cross-sectional average temperature located near the outlet, representing a measurement location relevant to the medical device usage.

Discretization error is assessed using a co-refined set of seven meshes (M1–M7) paired with progressively decreasing time steps. This design covers resolutions from coarse to very fine, enabling detailed assessment across the full discretization range. The meshes are topologically identical, geometrically similar, and globally refined with a refinement factor of $r = \sqrt[3]{2} \approx 1.26$, resulting in characteristic cell sizes of 0.37 mm (M1), 0.29 mm (M2), 0.23 mm (M3), 0.18 mm (M4), 0.14 mm (M5), 0.11 mm (M6), and 0.091 mm (M7). Each mesh is paired with a corresponding time step to maintain consistent co-refinement, with time steps of 0.004 s (T1), 0.0035 s (T2), 0.0025 s (T3), 0.002 s (T4), 0.0016 s (T5), 0.0014 s (T6), and 0.001 s (T7). The coarsest mesh (M1) is paired with the largest time step (T1), and the finest mesh (M7) with the smallest time step (T7), enabling assessment of convergence under simultaneous spatial and temporal refinement.

Resolution Methodology. Discretization error is assessed through systematic refinement of spatial and temporal resolution. Code verification studies employ spatial refinement only under steady-state conditions, while solution verification studies use a co-refined approach in which mesh resolution and time step are reduced simultaneously to evaluate convergence in the coupled transient system.

Solver Tolerance. Solver sensitivity is assessed by varying solver convergence tolerances over several orders of magnitude while holding spatial and temporal resolution fixed. The resulting variation in QoIs is used to quantify the influence of solver convergence criteria on numerical error.

Solver sensitivity is assessed by varying linear solver convergence tolerances over several orders of magnitude while holding spatial and temporal resolution fixed. Convergence tolerances of 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, and 10^{-8} are evaluated at M3 and time step 0.005 s. The resulting variation in QoIs is used to quantify the influence of solver convergence criteria on numerical error.

2.5.3 Decoupled Resolution Analysis. An additional analysis is performed to examine the impact of independently refining spatial and temporal resolution. Two cases are considered: (1) fixed spatial resolution with varying time step, and (2) fixed time step with varying spatial resolution. This study highlights the potential for misleading conclusions when discretization parameters are varied independently, demonstrating the importance of consistent refinement strategies in assessing convergence behavior. In Case 1, the spatial resolution is fixed at mesh M3 while the time step is systematically reduced using a refinement factor of $\sqrt{2}$. Ten time steps are evaluated: 0.0060 s, 0.0042 s, 0.0030 s, 0.0021 s, 0.0015 s, 0.0011 s, 0.0007 s, 0.0005 s, 0.0004 s, and 0.0003 s. In Case 2, the time step is fixed at 0.005 s while spatial resolution is varied using the same set of seven meshes (M1-M7) defined in the solution verification studies.

2.6 Analytical Solutions. Analytical solutions are used in code verification to assess whether the numerical implementation correctly represents the governing equations. Two problems are considered: fully developed laminar pipe flow and steady-state radial conduction with internal heat generation.

2.6.1 Hagen-Poiseuille Flow. The analytical solution for the axial velocity profile of fully developed laminar flow in a circular pipe is given by

$$u(r) = \frac{\Delta P}{4\mu L}\left(R^2 - r^2\right) \tag{8}$$

where u is the axial velocity as a function of radial position r, ΔP is the pressure drop along the pipe of length L, μ is the dynamic viscosity, and R is the pipe radius [23, 24].

2.6.2 Radial Conduction with Internal Heat Generation. The solid casing is modeled as a cylindrical annulus with inner radius r_{in} and outer radius r_w. The analytical solution assumes steady-state conditions, one-dimensional radial heat conduction, constant thermal conductivity k, uniform volumetric heat generation S, negligible axial conduction, and an adiabatic inner boundary at $r = r_{in}$.

Under these assumptions, the governing equation for steady radial conduction in cylindrical coordinates is

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) + \frac{S}{k} = 0. \tag{9}$$

Integrating twice and applying the boundary conditions yields the analytical temperature distribution:

$$T(r) = T_w - \frac{S}{4k}(r^2 - r_w^2) + \frac{Sr_{in}^2}{2k}\ln\left(\frac{r}{r_w}\right) \tag{10}$$

where $T(r)$ is the temperature as a function of radial position, S is the uniform volumetric heat source, k is the thermal conductivity, r_{in} is the inner radius at the solid-fluid interface, r_w is the outer wall radius, and T_w is the prescribed temperature at the outer wall.

2.6.3 Transient Thermal Diffusion Model. The transient thermal diffusion model assumes zero fluid velocity, reducing the energy equation to a radial diffusion problem in cylindrical coordinates. The resulting governing equation is

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)\right) \tag{11}$$

where T is temperature, t is time, r is the radial coordinate, and α is the thermal diffusivity. For a cylinder the solution is written in terms of the dimensionless temperature

$$\theta(r,t) = \frac{T(r,t) - T_{\text{solid}}}{T_0 - T_{\text{solid}}} \tag{12}$$

where T_0 is the initial temperature and T_{solid} is the temperature at the wall (fluid-solid interface). The solid has a constant temperature $T_0 + 10°C$. Applying separation of variables gives

$$\theta(r,t) = \sum_{n=1}^{\infty}\frac{2J_0\lambda_n r}{J_1 R\lambda_n^2}\exp\left(-\lambda_n^2\frac{\alpha t}{R^2}\right) \tag{13}$$

where J_0 and J_1 are Bessel functions of the first kind of order zero and one, respectively, R is the pipe radius, and λ_n are the positive roots of $J_0(\lambda_n) = 0$ associated with the wall temperature boundary condition. The dimensional analytical temperature is then recovered as

$$T(r,t) = T_{\text{solid}} + \theta(r,t)\,(T_0 - T_{\text{solid}}) \tag{14}$$

The analytical solution was evaluated using a truncated series expansion with $n = 100$ terms. Additional sensitivity studies were performed for truncated expansions ranging from $n = 5$ to $n = 5000$ where errors were negligible beyond approximately $n = 25$.

While the analytical benchmark problems represent simplified physical systems, they were selected to isolate key phenomena present in the full model and provide rigorous verification targets with known solutions. Agreement with these benchmarks increases confidence in the numerical implementation of the governing equations and supports subsequent application of the model to more complex configurations. Although analytical solutions are limited to simplified representations of the underlying physics, they remain valuable verification tools for building credibility.

2.7 Extrapolation Method. The `StREEQ` method is used to quantify discretization error and evaluate the convergence behavior of the numerical solution [25, 26]. The method estimates the observed order of convergence and extrapolates the solution to the zero-discretization limit. When numerical error is influenced by both spatial and temporal resolution, a regression-based model is used to capture their combined contributions:

$$QOI(h,\Delta t) = \beta_0 + \beta_1 h^{p_h} + \beta_2 \Delta t^{p_t} \tag{15}$$

where $QOI(h,\Delta t)$ is the quantity of interest as a function of representative mesh size h and time step Δt, β_0 is the extrapolated solution at zero discretization, β_1 and β_2 are regression coefficients, and p_h and p_t are the observed orders of convergence associated with spatial and temporal refinement, respectively. This formulation does not require a constant mesh refinement factor. When only a single source of discretization error is present, the model reduces to a single-parameter regression in either h or Δt.

2.8 Reproducibility and Computational Workflow. Reproducibility is treated as a foundational component of model credibility in this study. Verification activities rely on consistent and traceable computational processes to ensure that observed numerical behavior reflects controlled changes in resolution rather than unintended variability in model setup or execution. All simulations and post-processing steps were performed using scripted workflows and publicly available tools to enforce consistency across verification studies. Meshes were generated using *Cubit*® version 17.04 via journal scripts to ensure consistency across refinements. The resulting Fluent-format `.msh` files were converted to OpenFOAM format using the `fluentMeshToFoam` utility. Simulations were conducted using OpenFOAM version 9 for compatibility with the SOLO high-performance computing cluster at Sandia National Laboratories [27]. Jobs were submitted using the SLURM scheduler and automated via `sbatch` scripts and Dakota 6.20 [28], coordinating mesh decomposition (`decomposePar`), parallel execution (`mpirun`), and solution reconstruction (`reconstructPar`). Simulations were executed on multi-core CPU nodes, with runtimes ranging from under two minutes for coarse configurations to over two days for the most refined cases. Post-processing was performed using Python 3, with analysis and visualization routines implemented using NumPy, pandas, matplotlib, and PyVista. Simulation outputs were converted from VTK format to structured `.csv` files to enable consistent extraction and analysis of quantities of interest. All computational artifacts, including mesh files, solver configurations, workflow scripts, and post-processing tools, are publicly available via a GitHub repository and archived datasets to support transparency, reproducibility, and independent verification [29]. The GitHub repository can be found in https://github.com/plinarodriguez/edds_verification_repo.

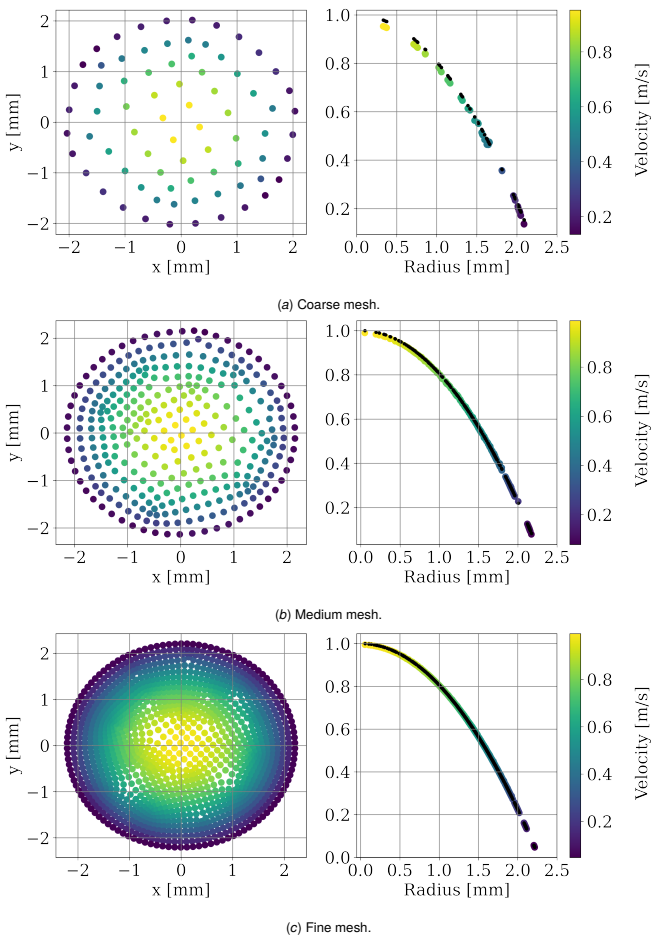
3 Results

This section presents the results of the verification studies used to assess the numerical accuracy and convergence behavior of the computational model. The analysis begins with steady-state code verification, where numerical solutions are compared against analytical benchmarks for fluid flow and heat conduction. This is followed by solution verification studies, including both steady-state and transient configurations, to quantify discretization error arising from spatial and temporal resolution, as well as solver tolerance. Together, these results provide evidence of numerical correctness, convergence, and overall model credibility.

3.1 Code Verification. Code verification is performed to assess the correctness of the numerical implementation through comparison with analytical solutions. Three problems are considered to independently verify the fluid and thermal components of the `chtMultiRegionFoam` solver: Hagen–Poiseuille flow for laminar

momentum transport, steady-state radial conduction with internal heat generation [5, 23], and transient thermal diffusion using a separation-of-variables Bessel-function solution. The transient diffusion configuration additionally verifies the transient thermal coupling behavior at the fluid-solid interface by ensuring that the imposed solid temperature response is accurately transferred into the fluid region. Although the full conjugate heat transfer solver is used, the configurations are constructed to isolate fluid flow and heat conduction, respectively. This approach enables rigorous verification of each governing equation prior to evaluating the fully coupled system under solution verification.

3.1.1 Hydrodynamic (Fluid Only) Model. To verify the momentum transport implementation, a steady-state laminar pipe flow simulation was compared against the analytical Hagen-Poiseuille solution. Spatial convergence was assessed using three progressively refined meshes with representative cell sizes of 0.43 mm, 0.22 mm, and 0.11 mm, while maintaining identical geometry and boundary conditions. Axial velocity was evaluated at a cross-sectional plane located beyond the hydrodynamic entry length to ensure fully developed flow. The numerical velocity field was compared pointwise with the analytical solution (Eq. 8). Figure 2 shows that the numerical solution progressively aligns with the analytical parabolic profile as mesh resolution improves. The coarse mesh overpredicts velocity near the centerline due to insufficient spatial resolution, while refinement at the finest mesh achieves the best agreement.



(*a*) Coarse mesh.

(*b*) Medium mesh.

(*c*) Fine mesh.

Fig. 2 Simulated hydrodynamic model axial velocity by radial position at QoI_{hydro}, overlaid with the analytical Hagen-Poiseuille solution (black points).

The normalized L_2 error was evaluated for each mesh and plotted against representative cell length (Figure 3). The error decreases consistently with mesh refinement, yielding an observed order of convergence of $p = 1.95$. This behavior is consistent with the expected second-order accuracy of the spatial discretization scheme. These results confirm that the numerical solver accurately represents the analytical velocity field.

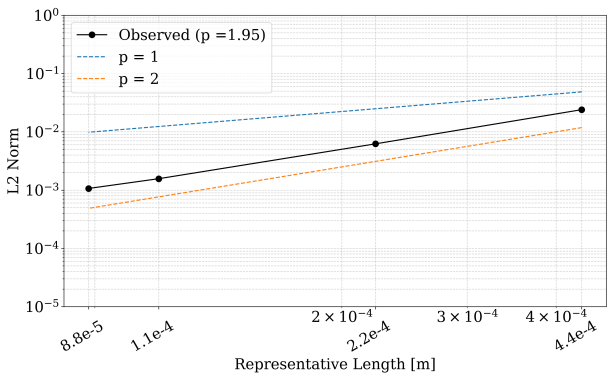


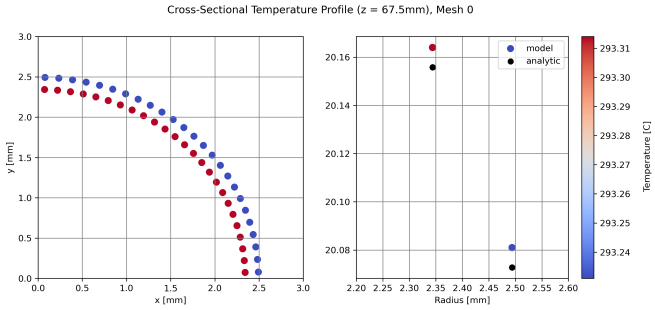
Fig. 3 Hydrodynamic model velocity L_2 error by representative cell length.

3.1.2 Radial Conduction Model. To verify the implementation of the solid energy equation, a steady-state conduction-dominated configuration was constructed with the `chtMultiRegionFoam` solver. The problem was formulated such that the governing equations reduce to a one-dimensional radial conduction with uniform volumetric heat generation in the solid, enabling direct comparison with the analytical solution. Spatial convergence was assessed using three progressively refined meshes with representative cell sizes of 0.14 mm, 0.071 mm, and 0.035 mm. Temperature was evaluated at a cross-sectional plane sufficiently far from inlet and outlet effects to ensure a fully developed radial profile, presented in Figure 4.

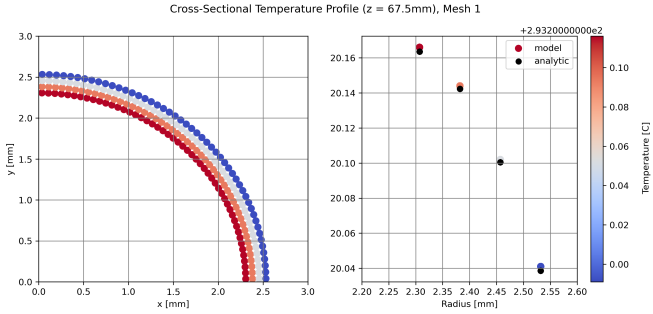
Figure 4 shows that the numerical temperature field converges toward the analytical solution as mesh resolution increases. The coarse mesh exhibits noticeable deviation from the analytical profile, while refinement reduces this discrepancy and the fine mesh achieves near pointwise agreement across the radius. The normalized L_2 error was evaluated for each mesh and plotted against representative cell length (Figure 5). The error decreases consistently with mesh refinement, producing an observed order of convergence of $p = 2.08$, which is consistent with the expected second-order spatial accuracy of the discretization scheme. The error magnitude decreases from $\mathcal{O}(10^{-4})$ to $\mathcal{O}(10^{-6})$, indicating strong agreement with the analytical solution. These results confirm that the numerical implementation of the solid energy equation is correct and exhibits the expected convergence behavior under mesh refinement.

3.1.3 Transient Thermal Diffusion Verification. Following isolated verification of the hydrodynamic and steady-state radial conduction models, a transient thermal diffusion case is used to bridge the isolated verification studies and the fully coupled transient conjugate heat transfer model. In order to reduce the governing energy equation to a radial diffusion problem the fluid velocity was set to zero. This isolates transient thermal transport and interface temperature transfer from the solid to the fluid region. The L_2 error is computed at a cross section, halfway along the heated fluid region, by comparing the simulated temperature, at each radial position, to the corresponding analytical radial temperature obtained from Eq. 13 and Eq. 14. Figure 6 shows the normalized L_2 error as a function of representative mesh length for multiple transient times.

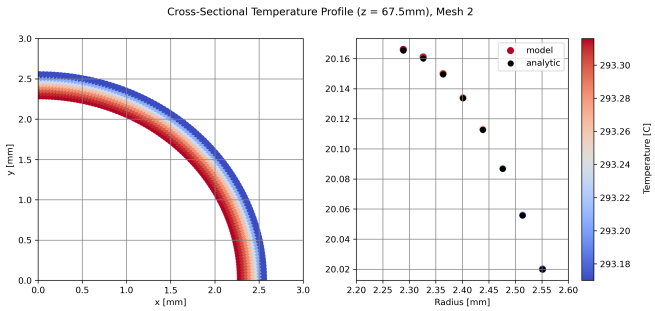
The error decreases consistently with mesh refinement, yielding observed orders of convergence of $p = 2.47$ at $t = 0.015$ s, $p = 2.30$ at $t = 0.020$ s, and $p = 2.15$ at $t = 0.025$ s, showing near second-order spatial accuracy. Larger errors occur at earlier times due to stronger thermal gradients, while later times exhibit reduced error magnitudes as the solution becomes increasingly diffused. These



(*a*) Coarse mesh.



(*b*) Medium mesh.



(*c*) Fine mesh.

Fig. 4 Radial conduction model simulated temperature by radial position at QoI_{cond}, overlaid with the analytical solution (black points).

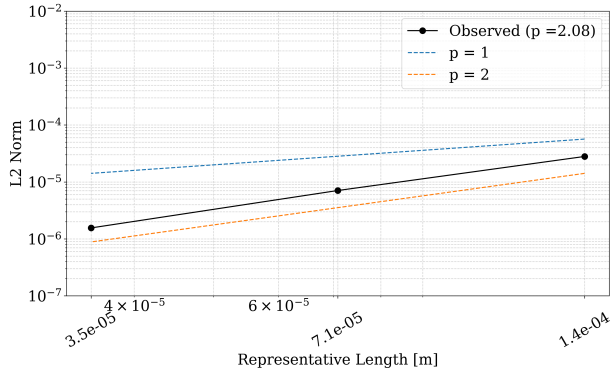


Fig. 5 Radial conduction model temperature L_2 error by representative cell length.

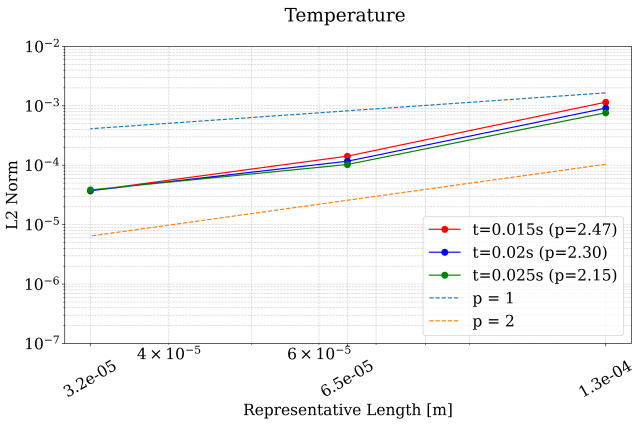


Fig. 6 Thermal diffusion model temperature L_2 error by representative cell length at three times.

results demonstrate that the solver accurately captures transient thermal diffusion behavior and correctly transfers the imposed solid temperature response across the fluid-solid interface under transient conditions.

3.2 Solution Verification. Solution verification results are presented to quantify numerical error arising from spatial and temporal discretization, as well as solver convergence. The analysis focuses on the impact of mesh resolution, time step size, and solver tolerance on the predicted temperature response. The following studies evaluate convergence behavior and discretization error for both steady-state and transient models, building the accuracy and reliability of the coupled fluid-thermal solution.

3.2.1 Steady-State Coupled Pipe. Following independent verification of the fluid and solid governing equations, a fully coupled conjugate heat transfer (CHT) configuration is considered to assess the combined system behavior. The discretization error is evaluated through systematic mesh refinement under steady-state conditions. Figure 7 shows the radial temperature distribution within the thermally developing region, located 14 mm downstream of the start of heating. The highest temperatures occur near the wall where heat is introduced, while lower temperatures are near the centerline due to the developing thermal boundary layer. As mesh resolution increases, the temperature field becomes smoother and more consistent.

To quantify convergence, the bulk fluid temperature was evaluated downstream of the thermal development region at 30 mm and compared across mesh resolutions using an extrapolated solution obtained from the `StREEQ` method. Figure 8 shows the relative error as a function of representative mesh length on a log-log scale. The error decreases consistently with refinement, resulting in an observed order of convergence of $p = 1.90$, which is consistent with near second-order spatial accuracy. These results demonstrate that the coupled solver exhibits consistent convergence behavior and accurately captures the thermally developing flow.

3.2.2 Transient Simplified EDDS. To evaluate the coupled effects of spatial and temporal discretization, a co-refinement study was performed in which seven mesh levels (M1–M7) were paired with progressively refined time steps (T1–T7). This approach assesses how discretization errors evolve in a transient, coupled fluid-thermal system. Figure 9 shows the relative error estimated using the `StREEQ` model (Eq. 15) across all mesh-timestep combinations. The error ranges from $\mathcal{O}(10^{-2})$ for coarse configurations to $\mathcal{O}(10^{-4})$ for the finest cases. A trend is observed in which error decreases with both mesh refinement and timestep reduction, indicating that spatial and temporal discretization contribute jointly to the numerical error. The relative errors show a nearly linear

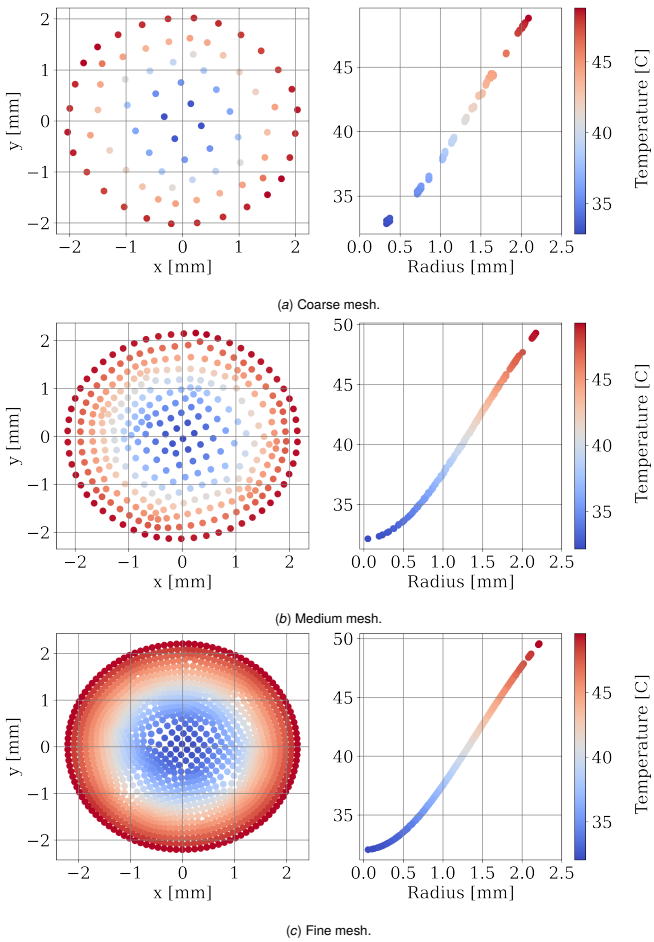


Fig. 7 Radial temperature distribution within the thermally developing region.

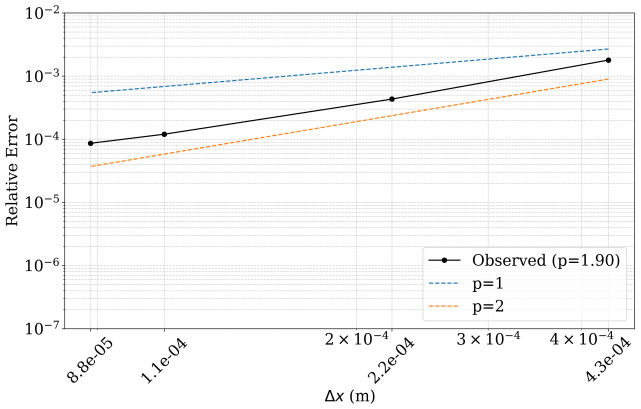


Fig. 8 Relative error in bulk temperature versus representative mesh length, with observed order of convergence $p = 1.90$.

gradient across the mesh-timestep space, suggesting a balanced influence of spatial and temporal resolution. This behavior highlights the coupled nature of discretization error in transient simulations, where neither spatial nor temporal refinement alone is sufficient to achieve convergence.

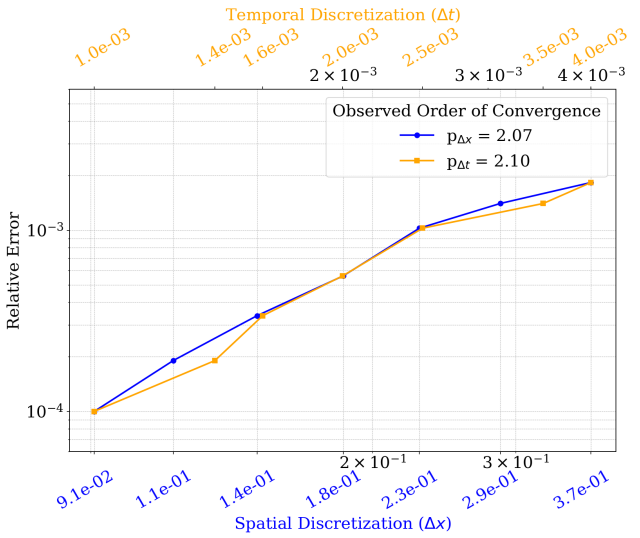


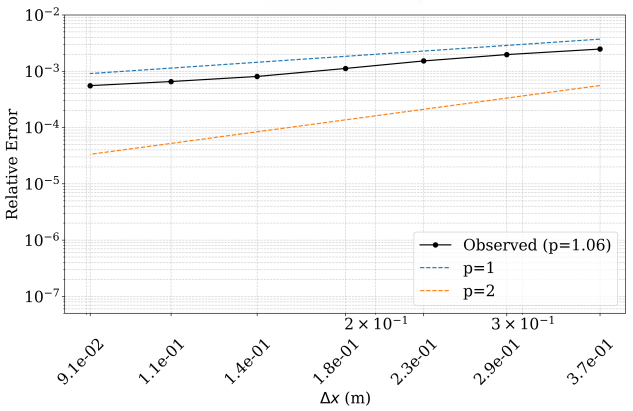
Fig. 9 Relative error across mesh-timestep combinations from the co-refinement study.

3.2.3 Solver Tolerance Study. This study is conducted for the transient simplified EDDS configuration, where iterative solver convergence can influence the time-dependent thermal response. To assess the impact of linear solver convergence criteria on the predicted thermal response, a solver tolerance study was conducted using mesh M3 and time step T3. Six convergence tolerances ranging from 10^{-3} to 10^{-8} were evaluated. The coarsest tolerance (10^{-3}) underpredicts the temperature response throughout the simulation. As the tolerance is tightened, the solutions converge toward a consistent response. The results for 10^{-7} and 10^{-8} are visually indistinguishable, indicating that solver-induced numerical error is negligible at these tolerances. Relative error was computed using the most stringent tolerance (10^{-8}) as the reference solution. The error decreases from $\mathcal{O}(10^{-1})$ at 10^{-3} to $\mathcal{O}(10^{-4})$ at 10^{-8}. A tolerance of 10^{-6} yields a relative error below 0.0125%, indicating sufficient solver accuracy for this problem.

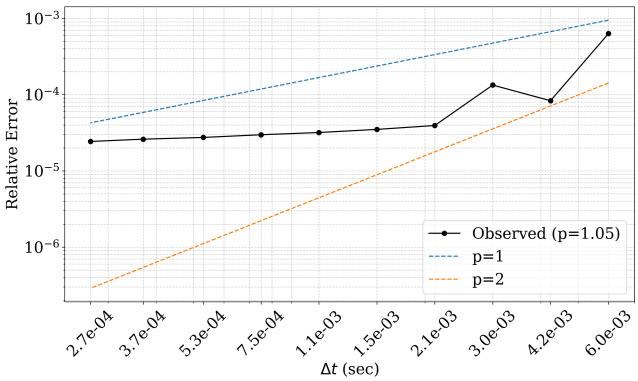
3.2.4 Decoupled Resolution Analysis. To assess the impact of independently refining spatial and temporal discretization, mesh-only and timestep-only studies were performed and compared against the co-refinement results. Figure 10 shows that while error decreases with refinement in both cases, the observed convergence rates ($p \approx 1.06$) are significantly lower than the expected second-order behavior. This indicates that unresolved discretization error from the fixed parameter (time or space) influences the solution.

Comparison with the co-refinement results (Figure 11) shows that independent refinement can under- or overestimate numerical error. In contrast, co-refinement captures the coupled influence of spatial and temporal discretization, providing a more reliable assessment of convergence behavior. These results demonstrate that independent refinement is insufficient for accurately characterizing discretization error in transient conjugate heat transfer simulations.

3.3 Interface Energy Consistency. To ensure that the coupled conjugate heat transfer satisfies heat-flux continuity, an additional analysis was performed on the simplified EDDS model at the fluid-solid interface with the interface discretization refining consistently with the global mesh. It is expected that the heat flux



(*a*) Mesh-only refinement (fixed timestep).



(*b*) Timestep-only refinement (fixed mesh).

Fig. 10 Observed convergence behavior for independent mesh and timestep refinement studies. Both cases exhibit reduced convergence rates ($p \approx 1.06$), indicating the influence of unresolved discretization error from the fixed parameter.

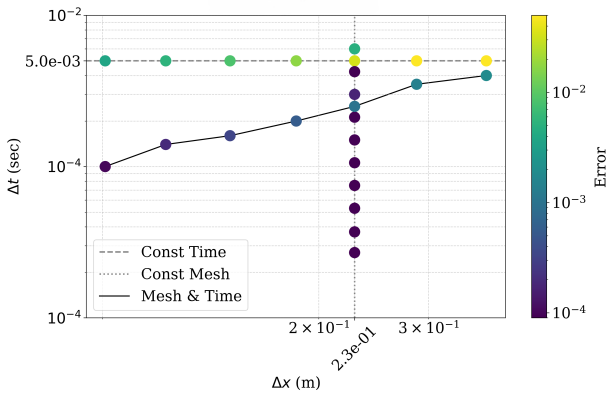


Fig. 11 Relative error across mesh and timestep combinations, including co-refined and independently refined cases.

leaving the solid must equal the heat flux entering the fluid, such that $Q_{solid} + Q_{fluid} \approx 0$. The interface imbalance was defined as:

$$\epsilon_{iface} = |Q_{solid} + Q_{fluid}|.$$

The interface energy imbalance, normalized by the applied heat source, remained on the order of $\mathcal{O}(10^{-4})$ across all mesh resolutions and did not exhibit a clear dependence on mesh refinement. Multiple competing errors, which include interface interpolation, solver tolerances, and velocity-temperature coupling, prevent the interface imbalance from decreasing with mesh refinement because the interface flux depends on temperature gradients that are highly sensitive to local discretization.

3.4 Computational Cost. Computational cost was evaluated for the transient simplified EDDS configuration to assess the trade-off between numerical accuracy and resource requirements. Figure 12 summarizes the cost across mesh and timestep combinations. The simulations for each of these combinations were conducted on an Intel Xeon Broadwell (BDW) E5-2695v4 at 2.1 GHz with 18 core CPU workstations.

For the intermediate resolution case (M3 with $\Delta t = 0.0025$ s), the relative error is approximately 0.1% with a computational cost of 24 minutes. Refinement to M4 with $\Delta t = 0.002$ s reduces the error to 0.055%, but increases computational cost to 47 minutes, corresponding to a 96% increase in cost for a 50% reduction in error.

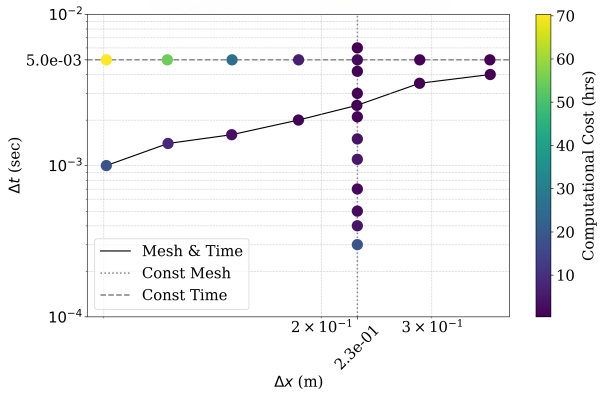


Fig. 12 Computational Cost in hours across mesh, time step, and combined mesh time step studies.

These results demonstrate diminishing returns with increased resolution, where significant increases in computational cost yield comparatively small gains in accuracy. This trade-off highlights the importance of selecting discretization parameters that balance accuracy with computational efficiency, particularly in resource-constrained environments. Based on this analysis, the M3–T3 configuration provides a practical balance between accuracy and cost, and is selected for subsequent validation studies. This selection also supports reproducibility by enabling the study to be executed within accessible computational resources while maintaining sufficient numerical accuracy.

4 Discussion

This study presents a rigorous verification process for a conjugate heat transfer model of a simplified medical device geometry. The objective was to establish confidence in the numerical implementation and to quantify sensitivity to spatial and temporal discretization, as well as solver convergence. These verification activities form a critical step toward building model credibility prior to validation.

Verification plays a particularly important role in medical device modeling, where experimental validation is often limited by cost, feasibility, or measurement uncertainty. In such cases, verification provides a quantitative framework to isolate numerical error, ensuring that remaining uncertainty can be attributed to physical modeling assumptions or experimental limitations rather than deficiencies in the numerical solution.

Code verification results confirm that the numerical implementation accurately represents the governing equations. The hydrodynamic case exhibited an observed order of convergence of $p = 1.90$, with errors decreasing from $\mathcal{O}(10^{-3})$ to $\mathcal{O}(10^{-5})$, while the radial conduction case achieved $p = 2.08$, consistent with second-order spatial accuracy. These results demonstrate that both the momentum and energy equations are correctly discretized and solved under idealized conditions.

The verification process was intentionally structured to progressively isolate the governing physics relevant to the final coupled transient configuration. The hydrodynamic verification case isolated the momentum equations and confirmed accurate velocity-field discretization independent of thermal effects. The radial conduction case then isolated steady-state thermal transport within the solid domain by removing fluid effects through an adiabatic interface boundary condition. While these studies independently verified the momentum and thermal governing equations, neither configuration evaluated transient thermal transport or the thermal coupling behavior at the fluid-solid interface. To bridge this gap, a transient thermal diffusion verification case was introduced by imposing zero velocity throughout the fluid region, reducing the governing energy equation to a pure diffusion problem with an analytical Bessel-function solution. This intermediate verification configuration enabled assessment of transient thermal transport behavior independently of advection while also verifying that the transient solid temperature response was correctly imposed across the fluid-solid interface into the fluid region. The transient diffusion study demonstrated near second-order spatial convergence, with observed orders ranging from $p = 2.15$ to $p = 2.47$ across multiple transient times.

Solution verification results highlight the importance of evaluating discretization error in coupled, application-relevant configurations. In the transient simplified EDDS case, co-refined mesh and timestep studies revealed that spatial and temporal discretization errors interact and accumulate over time. Independent refinement of either parameter produced reduced convergence rates ($p \approx 1.06$), indicating that unresolved error from the fixed parameter influences the solution. This demonstrates that accurate error estimation in transient multiphysics simulations requires consistent co-refinement of spatial and temporal discretization.

The interface energy consistency results support the coupled solution. The interface heat flux is more sensitive to local discretization since it does not exhibit monotonic convergence with mesh refinement. The imbalance remains small, approximately 0.01% of the applied heat source, indicating that energy is well conserved across the interface. This imbalance is reasonable so long as it remains smaller than the discretization error.

The computational cost analysis further emphasizes the practical implications of verification. While refinement improves accuracy, diminishing returns are observed as computational cost increases. Importantly, the relationship between discretization parameters and computational cost is not strictly monotonic. Larger timesteps or coarser meshes do not always yield lower runtime due to increased solver iterations and numerical stiffness. This highlights the need to evaluate computational cost based on measured runtime rather than assuming a direct relationship with discretization parameters.

The selected M3–T3 configuration provides a balance between accuracy (relative error $\sim 0.1\%$) and computational cost (24 minutes), making it suitable for downstream validation studies. This choice reflects a risk-informed approach to modeling, where sufficient accuracy is achieved without exceeding practical computational constraints. If stricter error tolerances are required for a given application, further refinement can be applied; however, this will incur increased computational cost and should be justified based on the sensitivity of the decision-making context.

Geometric simplification was essential to enable rigorous verification while retaining the dominant physics of the system. Although detailed device features such as coil geometry and the expanding mouthpiece were excluded, the simplified model captures the key fluid and thermal transport behavior governing the system response. This level of abstraction allows verification to be performed under controlled conditions while maintaining relevance to the physical system. More complex geometric features can be reintroduced during validation without compromising the credibility established through verification.

Finally, this study highlights the role of reproducibility in verification. High-fidelity simulations require large datasets and substantial computational resources, particularly for fine mesh and timestep combinations. By implementing scripted workflows and providing publicly accessible computational artifacts, this work ensures that verification results are traceable, reproducible, and accessible. This is particularly important in settings where computational resources are limited, as it enables broader use and independent assessment of the modeling approach. This verification process establishes a credible foundation for subsequent validation and application of the model within the ASME V&V 40 framework.

This work is intentionally scoped to verification in order to isolate and quantify numerical error prior to introducing additional sources of uncertainty. While uncertainty quantification and validation are essential components of the full VVUQ process, combining all elements within a single study can obscure the attribution of error sources. By separating verification from validation and uncertainty quantification, this work establishes a controlled and transparent foundation for subsequent analysis. A companion study will extend this work to include validation and uncertainty quantification under application-relevant conditions.

5 Conclusion

This study represents the first step in a comprehensive credibility assessment for a computational model of a simplified medical device geometry involving conjugate heat transfer. Using `OpenFOAM`'s `chtMultiRegionFoam` solver, a rigorous verification effort was conducted, including both code and solution verification, to evaluate numerical accuracy and implementation.

The results demonstrate that the solver reliably captures laminar flow and heat transfer behavior, with discretization errors from spatial resolution, timestep size, and solver tolerance systematically reduced to acceptable levels. Importantly, this study shows that the choice of resolution directly impacts both numerical accuracy and computational cost, requiring a careful balance between the two. A key finding is that spatial and temporal discretization errors in transient coupled simulations cannot be assessed independently. Co-refinement of mesh and timestep is necessary to obtain reliable estimates of numerical error, while independent refinement can lead to misleading convergence behavior. This highlights the importance of consistent and systematic verification practices when evaluating multiphysics models. The use of a simplified geometry enabled rigorous verification while preserving the dominant fluid and thermal transport physics of the device. This approach demonstrates that meaningful verification can be achieved even when analytical solutions are limited and full system complexity is not yet incorporated.

The interface energy consistency analysis further supports the coupled solution. The interface imbalance did not exhibit monotonic convergence with mesh refinement, it remained small, approximately $\mathcal{O}(10^{-4})$ of the applied heat source, thus the energy is well conserved across the fluid–solid interface and that the observed convergence behavior is not associated with a non-physical coupling error. This work also emphasizes practical considerations in verification, including computational cost and reproducibility.

High-fidelity simulations require significant resources and generate large datasets, particularly for fine mesh and timestep combinations. By identifying an efficient resolution strategy and providing reproducible workflows and publicly accessible artifacts, this study supports broader accessibility and transparency in computational modeling.

This study is aligned with the ASME V&V 40 framework and contributes a reproducible, risk-informed case study for building model credibility in medical device applications. For the medium–low risk context considered here, the verification approach demonstrates that a level of rigor beyond minimal practice can be achieved through targeted model simplification and systematic analysis. Establishing numerical correctness and quantifying discretization error provides a critical foundation for subsequent validation and uncertainty quantification efforts. Future work will build upon this verification foundation by incorporating validation and uncertainty quantification to complete the full VVUQ framework. Overall, this work demonstrates that systematic verification of coupled fluid–thermal models, even in the absence of analytical solutions for the full system, can provide credible and decision-relevant modeling evidence under practical computational constraints.

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This paper describes objective technical results and analysis. Any subjective views or opinions that might be expressed in the paper do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

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